Federico

Section 1 — Auth & Users

1. ROLE

Schema

```
ROLE (
role_id INT PRIMARY KEY,
role_name VARCHAR(50) UNIQUE NOT NULL
)
```

Relationship

ROLE (1) —— (N) USER

Participation

USER → total participation

ROLE → partial participation

Meaning: every user must have a role.

role_id	role_name
1	Admin
2	Patient

---

2. USER

Schema

```
USER (
user_id INT PRIMARY KEY,
name VARCHAR(100),
email VARCHAR(100) UNIQUE,
password_hash VARCHAR(255),
role_id INT NOT NULL,
created_at TIMESTAMP,
FOREIGN KEY (role_id) REFERENCES ROLE(role_id)
)
```

Relationships

USER (1) —— (0..1) PATIENT

USER (1) —— (N) PURCHASE_REQUEST

Participation

PATIENT → total

USER → partial

user_id	name	email	password_hash	role_id	created_at
101	Sarah Connor	sarah.c@hosp.com	$2b$12$eJ...	1	2026-03-01 10:00:00
102	John Doe	john.doe@gmail.com	$2b$12$kP...	2	2026-03-02 11:30:00

Section 2 — Patients

3. PATIENT

Schema

```
PATIENT (
patient_id INT PRIMARY KEY,
user_id INT NOT NULL,
uhid VARCHAR(20) UNIQUE NOT NULL,
name VARCHAR(100) NOT NULL,
phone VARCHAR(15),
alternate_phone VARCHAR(15),
dob DATE,
gender VARCHAR(10),
blood_group VARCHAR(5),
address TEXT,
emergency_contact_name VARCHAR(100),
emergency_contact_phone VARCHAR(15),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES USER(user_id)
)
```

Relationships

PATIENT (1) —— (N) APPOINTMENT

PATIENT (1) —— (N) ADMISSION

PATIENT (1) —— (N) PATIENT_INSURANCE

PATIENT (1) —— (N) DISCHARGE_SUMMARY

Participation

PATIENT → partial participation

APPOINTMENT → total participation

patient_id	user_id	uhid	name	phone	alternate_phone	dob	gender
201	102	UHID-882100	John Doe	+91-9876543210	+91-9876543211	1990-05-15	Male
202	103	UHID-994200	Jane Smith	+91-9123456789	NULL	1985-11-20	Female

---

4. PATIENT_INSURANCE

Schema

```
PATIENT_INSURANCE (
insurance_id INT PRIMARY KEY,
patient_id INT NOT NULL,
provider_name VARCHAR(100),
policy_number VARCHAR(50),
member_id VARCHAR(50),
coverage_type VARCHAR(20),
valid_from DATE,
valid_to DATE,
```

```
created_at TIMESTAMP,
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id)
)
```

Relationships

PATIENT (1) —— (N) PATIENT_INSURANCE

PATIENT_INSURANCE (1) —— (0..N) PATIENT_INSURANCE_DOCUMENTS

Participation

PATIENT_INSURANCE → total

PATIENT → partial

insurance_id	patient_id	provider_name	policy_number	member_id	coverage_type	valid_from	valid_to
301	201	BlueCross	BC-77210	M-990	Full	2025-01-01	2027-12-31
302	202	Star Health	SH-99102	M-112	Basic	2026-01-01	2028-06-30

5. PATIENT_INSURANCE_DOCUMENTS

Schema

```
PATIENT_INSURANCE_DOCUMENTS (
doc_id INT PRIMARY KEY,
insurance_id INT NOT NULL,
file_name VARCHAR(255),
file_path VARCHAR(500),
uploaded_at TIMESTAMP,
FOREIGN KEY (insurance_id) REFERENCES PATIENT_INSURANCE(insurance_id)
)
```

Relationship

PATIENT_INSURANCE (1) —— (N) PATIENT_INSURANCE_DOCUMENTS

Participation

PATIENT_INSURANCE_DOCUMENTS → total

PATIENT_INSURANCE → partial

doc_id	insurance_id	file_name	file_path	uploaded_at
1	301	policy_copy.pdf	/docs/ins/301/p1.pdf	2026-03-02 12:00:00
2	302	id_card.jpg	/docs/ins/302/id.jpg	2026-03-03 10:00:00

Section 3 — Doctors

6. DOCTOR

Schema

```
DOCTOR (
doctor_id INT PRIMARY KEY,
name VARCHAR(100),
specialization VARCHAR(100),
phone VARCHAR(15),
```

```
email VARCHAR(100)
)
```

Relationship

DOCTOR (1) —— (N) DOCTOR_AVAILABILITY

Participation

DOCTOR_AVAILABILITY → total

DOCTOR → partial

doctor_id	name	specialization	phone	email
401	Dr. Aris Thorne	Cardiology	8881112222	thorne@hosp.com
402	Dr. Elena Vance	Neurology	8883334444	vance@hosp.com

---

7. DOCTOR_AVAILABILITY

Schema

```
DOCTOR_AVAILABILITY (
availability_id INT PRIMARY KEY,
doctor_id INT NOT NULL,
available_date DATE,
start_time TIME,
end_time TIME,
status VARCHAR(20),
FOREIGN KEY (doctor_id) REFERENCES DOCTOR(doctor_id)
)
```

Relationships

DOCTOR (1) —— (N) DOCTOR_AVAILABILITY

DOCTOR_AVAILABILITY (1) —— (0..N) APPOINTMENT

availability_id	doctor_id	available_date	start_time	end_time	status
501	401	2026-03-15	09:00:00	12:00:00	Available
502	402	2026-03-15	14:00:00	17:00:00	Available

---

8. APPOINTMENT

Schema

```
APPOINTMENT (
appointment_id INT PRIMARY KEY,
patient_id INT NOT NULL,
availability_id INT NOT NULL,
scheduled_datetime TIMESTAMP NOT NULL,
visit_type VARCHAR(20)
CHECK (visit_type IN ('OPD','EMERGENCY','FOLLOWUP')),
status VARCHAR(20)
CHECK (status IN ('PENDING','CONFIRMED','CANCELLED','COMPLETED')),
created_by INT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id),
```

```
FOREIGN KEY (availability_id) REFERENCES DOCTOR_AVAILABILITY(availability_id),
FOREIGN KEY (created_by) REFERENCES USER(user_id)
)
```

Relationships

PATIENT (1) —— (N) APPOINTMENT

DOCTOR_AVAILABILITY (1) —— (N) APPOINTMENT

APPOINTMENT (1) —— (0..1) ADMISSION

Participation

APPOINTMENT → total

PATIENT → partial

appointment_id	patient_id	availability_id	scheduled_datetime	visit_type	status	created_by	created_
601	201	501	2026-03-15 09:30:00	OPD	CONFIRMED	101	2026-03
602	202	502	2026-03-15 14:15:00	FOLLOWUP	COMPLETED	101	2026-03

Section 4 — Wards

9. WARD

Schema

```
WARD (
ward_id INT PRIMARY KEY,
ward_name VARCHAR(50) UNIQUE NOT NULL,
total_beds INT NOT NULL,
description TEXT
)
```

Relationship

WARD (1) —— (N) BED

Participation

BED → total

WARD → partial

ward_id	ward_name	total_beds	description
1	General Ward A	20	North Wing Floor 1
2	ICU - 01	10	Critical Care Unit

---

10. BED

Schema

```
BED (
bed_id INT PRIMARY KEY,
ward_id INT NOT NULL,
bed_number VARCHAR(10),
```

```
status VARCHAR(20),
FOREIGN KEY (ward_id) REFERENCES WARD(ward_id)
)
```

Relationships

WARD (1) —— (N) BED

BED (1) —— (N) ADMISSION

bed_id	ward_id	bed_number	status
11	1	G-101	OCCUPIED
22	2	ICU-05	AVAILABLE

---

11. ADMISSION

Schema

```
ADMISSION (
admission_id INT PRIMARY KEY,
appointment_id INT NOT NULL,
patient_id INT NOT NULL,
bed_id INT NOT NULL,
admit_time TIMESTAMP,
discharge_time TIMESTAMP,
status VARCHAR(20),
FOREIGN KEY (appointment_id) REFERENCES APPOINTMENT(appointment_id),
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id),
FOREIGN KEY (bed_id) REFERENCES BED(bed_id)
)
```

Relationships

APPOINTMENT (1) —— (0..1) ADMISSION

PATIENT (1) —— (N) ADMISSION

BED (1) —— (N) ADMISSION

ADMISSION (1) —— (1) LEDGER

ADMISSION (1) —— (0..1) INSURANCE

ADMISSION (1) —— (1) DISCHARGE_SUMMARY

Participation

ADMISSION → total

APPOINTMENT → partial

admission_id	appointment_id	patient_id	bed_id	admit_time	discharge_time	status
701	601	201	11	2026-03-15 10:30:00	2026-03-18 10:00:00	DISCHARGED
702	602	202	22	2026-03-16 08:00:00	NULL	ADMITTED

---

12. DISCHARGE_SUMMARY

Schema

```
DISCHARGE_SUMMARY (
summary_id INT PRIMARY KEY,
```

```
admission_id INT UNIQUE,
patient_id INT,
discharge_notes TEXT,
final_amount DECIMAL(10,2),
generated_at TIMESTAMP,
file_path VARCHAR(255),
FOREIGN KEY (admission_id) REFERENCES ADMISSION(admission_id),
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id)
)
```

Relationships

ADMISSION (1) —— (1) DISCHARGE_SUMMARY

PATIENT (1) —— (N) DISCHARGE_SUMMARY

summary_id	admission_id	patient_id	discharge_notes	final_amount	generated_at	file_path
1	701	201	Recovered well.	15500.00	2026-03-18	/path/ds_201.pdf
2	702	202	Followup in 7 days.	8200.00	2026-03-19	/path/ds_202.pdf

---

Section 5 — Billing

13. SERVICE

Schema

```
SERVICE (
service_id INT PRIMARY KEY,
service_name VARCHAR(100),
base_cost DECIMAL(10,2)
)
```

Relationships

SERVICE (1) —— (N) LEDGER_ENTRY

SERVICE (1) —— (N) INVENTORY_ITEM

service_id	service_name	base_cost
1	Consultation Fee	500.00
2	Room Rent	5000.00

---

14. LEDGER

Schema

```
LEDGER (
ledger_id INT PRIMARY KEY,
admission_id INT UNIQUE,
status VARCHAR(20),
created_at TIMESTAMP,
FOREIGN KEY (admission_id) REFERENCES ADMISSION(admission_id)
)
```

Relationships

ADMISSION (1) —— (1) LEDGER

LEDGER (1) —— (N) LEDGER_ENTRY

LEDGER (1) —— (N) PAYMENT

ledger_id	admission_id	status	created_at
801	701	CLOSED	2026-03-15
802	702	OPEN	2026-03-16

---

15. LEDGER_ENTRY (Weak Entity)

Schema

```
LEDGER_ENTRY (
entry_id INT,
ledger_id INT NOT NULL,
service_id INT,
quantity INT,
unit_price DECIMAL(10,2),
amount DECIMAL(10,2),
entry_time TIMESTAMP,
PRIMARY KEY (entry_id, ledger_id),
FOREIGN KEY (ledger_id) REFERENCES LEDGER(ledger_id),
FOREIGN KEY (service_id) REFERENCES SERVICE(service_id)
)
```

Relationships

LEDGER (1) —— (N) LEDGER_ENTRY

SERVICE (1) —— (N) LEDGER_ENTRY

Participation

LEDGER_ENTRY → total

LEDGER → partial

Owner entity: LEDGER

entry_id	ledger_id	service_id	quantity	unit_price	amount	entry_time
1	801	1	1	500.00	500.00	2026-03-15
2	801	2	3	5000.00	15000.00	2026-03-15

---

16. INSURANCE

Schema

```
INSURANCE (
insurance_id INT PRIMARY KEY,
admission_id INT UNIQUE NOT NULL,
patient_insurance_id INT NOT NULL,
coverage_percentage DECIMAL(5,2),
claim_status VARCHAR(20)
CHECK (claim_status IN ('PENDING','APPROVED','REJECTED')),
FOREIGN KEY (admission_id) REFERENCES ADMISSION(admission_id),
```

```
FOREIGN KEY (patient_insurance_id) REFERENCES PATIENT_INSURANCE(insurance_id)
)
```

Relationship

ADMISSION (1) —— (0..1) INSURANCE

insurance_id	admission_id	patient_insurance_id	coverage_percentage	claim_status
1	701	301	80.00	APPROVED
2	702	302	50.00	PENDING

---

17. PAYMENT

Schema

```
PAYMENT (
payment_id INT PRIMARY KEY,
ledger_id INT NOT NULL,
amount_paid DECIMAL(10,2),
payment_mode VARCHAR(20),
payment_time TIMESTAMP,
FOREIGN KEY (ledger_id) REFERENCES LEDGER(ledger_id)
)
```

Relationship

LEDGER (1) —— (N) PAYMENT

payment_id	ledger_id	amount_paid	payment_mode	payment_time
901	801	3100.00	CASH	2026-03-18
902	802	1000.00	UPI	2026-03-16

Section 6 — Inventory

18. INVENTORY_ITEM

Schema

```
INVENTORY_ITEM (
item_id INT PRIMARY KEY,
item_name VARCHAR(100),
category VARCHAR(50),
stock_quantity INT,
reorder_level INT,
service_id INT,
FOREIGN KEY (service_id) REFERENCES SERVICE(service_id)
)
```

Relationships

INVENTORY_ITEM (1) —— (N) PURCHASE_REQUEST

SERVICE (1) —— (N) INVENTORY_ITEM

item_id	item_name	category	stock_quantity	reorder_level	service_id
10	Syringe 5ml	Consumable	500	100	NULL
20	Paracetamol	Medicine	1200	200	1

19. PURCHASE_REQUEST

Schema

```
PURCHASE_REQUEST (
request_id INT PRIMARY KEY,
item_id INT,
quantity_requested INT,
status VARCHAR(20),
requested_by INT,
requested_at TIMESTAMP,
FOREIGN KEY (item_id) REFERENCES INVENTORY_ITEM(item_id),
FOREIGN KEY (requested_by) REFERENCES USER(user_id)
)
```

Relationships

INVENTORY_ITEM (1) —— (N) PURCHASE_REQUEST

USER (1) —— (N) PURCHASE_REQUEST

request_id	item_id	quantity_requested	status	requested_by	requested_at
1001	10	1000	APPROVED	101	2026-03-05
1002	20	500	PENDING	101	2026-03-06